



Operating System

Complete Revision Notes

Tere Topics

1. OS ki Definition
2. Types of Operating System
3. Functions of OS
4. Services of OS
5. Linux Architecture
6. System Call
7. System Programs

1 Operating System (OS) kya hai?

आसान भाषा में

Operating System ek system software hai jo user aur computer hardware ke beech interface ka kaam karta hai.

Matlab user directly CPU, RAM, hard disk, keyboard etc. ko control nahi karta. OS user ke instructions ko hardware tak pahunchata hai.

Example: Hum computer mein bolte hain: "Meri file open karo."

USER → OS → HARDWARE → FILE

OS hardware se file ko read karwata hai aur screen par dikha deta hai.

Examples of OS

- Windows
- Linux
- Android
- macOS
- Unix

Exam mein definition: Operating System ek system software hai jo user aur computer hardware ke beech interface ka kaam karta hai तथा computer ke resources jaise CPU, memory, files aur input-output devices ko manage karta hai.

2 Types of Operating System

Operating System ko alag-alag types mein divide kiya gaya hai.

1 Batch Operating System

Isme similar jobs ko ek batch mein collect karke execute kiya jata hai. User ko har job ke execution ke time directly interact karne ki zarurat nahi hoti.

Example: Maan lo 100 students ke marks calculate karne hain. Pehle sabhi data collect hoga → ek batch banega → OS sabhi jobs ko process karega.

Main point: 📌 Jobs batch mein execute hoti hain.

2 Multiprogramming Operating System

Isme ek se zyada programs memory mein rakhe ja sakte hain. CPU ek program ke wait karne par doosre program ko execute kar sakta hai.

Example: Ek program disk se data read kar raha hai. CPU khaali baithne ke bajay doosre program ko execute kar sakta hai.

Fayda: CPU ka utilization badhta hai.

3 Multitasking Operating System

Isme user ek hi time par multiple tasks kar sakta hai.

Example: Computer par: Chrome chal raha hai, Music chal raha hai, Word mein document likh rahe ho. OS CPU ka time alag-alag tasks ko deta rehta hai.

📌 Isliye hume lagta hai ki sabhi kaam ek saath ho rahe hain.

4 Multi-user Operating System

Isme multiple users ek computer system ke resources ko use kar sakte hain.

Example: Linux/Unix server par multiple users simultaneously login kar sakte hain.

5 Multiprocessing Operating System

Isme computer mein ek se zyada processors/CPU cores ka use kiya jata hai.

Example: Agar system mein multiple CPU cores hain, to OS different tasks ko different cores par execute kar sakta hai.

Fayda: Performance aur processing speed improve hoti hai.

6 Real-Time Operating System (RTOS)

Isme system ko given time limit ke andar response dena hota hai. Yahan delay bahut important hota hai.

Example: Air traffic control, Medical monitoring systems, Industrial control systems.

Main point: 📌 Response fixed time ke andar milna chahiye.

7 Distributed Operating System

Isme multiple computers/networked systems milkar ek single system ki tarah kaam karte hain.

Simple example: Kai computers ke resources ko coordinate karke ek common task perform karna.

★ **Types yaad karne ka shortcut:** B M M M M R D

- B → Batch
- M → Multiprogramming
- M → Multitasking
- M → Multi-user
- M → Multiprocessing
- R → Real-Time
- D → Distributed

3 Functions of Operating System

Ye bahut important question hai. Test mein aa sakta hai.

Operating System ke main functions:

1. Process Management

OS computer mein chalne wale programs/processes ko manage karta hai.

Example: Chrome, VS Code aur music player ek saath chal rahe hain → OS in processes ko manage karta hai.

2. Memory Management

OS RAM/memory ko manage karta hai. Kaunsa program kitni memory use karega, memory kab allocate hogi aur kab free hogi — OS manage karta hai.

3. File Management

OS files aur folders ko manage karta hai.

Example: File create, File delete, File rename, File read/write, Folder manage.

4. Device Management

OS input/output devices ko control karta hai.

Example: Keyboard, Mouse, Printer, Monitor, Disk.

5. Security and Protection

OS unauthorized users ko system/resources access karne se rokta hai.

Example: Username + Password.

6. CPU Management

OS decide karta hai ki CPU kis process ko kab milega. Iske liye CPU scheduling techniques use hoti hain.

7. User Interface

OS user ko computer ke saath interact karne ka medium deta hai. Do common interfaces:

- GUI → Graphical User Interface
- CLI → Command Line Interface

8. Error Detection

OS system mein hone wali errors ko detect karta hai aur unhe handle karne mein help karta hai.

★ **Functions ko ek line mein yaad karo: Process + Memory + File + Device + Security + CPU + Interface + Error**

4 Services of Operating System

Ab Function aur Service mein difference samajh.

Function — OS andar se kya manage karta hai.

Service — OS user/program ko kya facilities provide karta hai.

Main OS Services

① Program Execution

OS program ko load → execute → terminate karne mein help karta hai.

Example: Tum Chrome open karte ho → OS Chrome ko memory mein load karke execute karta hai.

② I/O Operations

OS input/output devices ke saath communication provide karta hai.

Example: Keyboard se input lena aur monitor par output dikhana.

③ File-System Manipulation

OS user ko files/folders ke saath operations karne deta hai.

Example: Create, read, write, delete, rename.

④ Communication

Processes ko ek doosre ke saath communicate karne ki facility provide karta hai.

Example: Ek process doosre process ko data bhej sakta hai.

⑤ Error Detection

OS hardware/software mein hone wali errors ko detect karta hai.

⑥ Resource Allocation

OS available resources ko different programs/users ke beech allocate karta hai.

Example: CPU, RAM, disk etc.

⑦ Protection and Security

OS system ke resources ko unauthorized access se protect karta hai.

🔥 Function vs Service

Function	Service
----------	---------

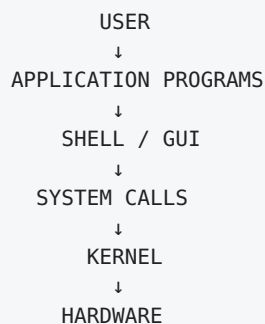
OS ka internal management	User/program ko facility
CPU manage karna	Program execute karwana
Memory manage karna	I/O operation provide karna
Process manage karna	File operation provide karna

Exam mein ye difference likh dena to answer strong lagega.

5 Linux Architecture

Ab sabse important diagram wala topic. 📌

Linux architecture ko simple layers mein samjho:



Ab ek-ek layer samajh.

① Hardware

Sabse neeche actual physical devices hote hain.

Example: CPU, RAM, Hard Disk, Keyboard, Mouse, Network devices.

② Kernel ★

Kernel Linux OS ka core/main part hota hai. Ye directly hardware ke saath communicate karta hai.

Kernel ke important kaam:

- Process management
- Memory management
- Device management
- File-system management
- Security

Simple example: Application ko CPU chahiye. Application directly CPU ko control nahi karegi.

APPLICATION → SYSTEM CALL → KERNEL → CPU

③ System Calls

System calls user program aur kernel ke beech interface provide karte hain. Program jab OS se koi service maangta hai, to system call ka use karta hai.

Example: File open karni hai → program OS ko request karega → system call → kernel file open karega.

④ Shell

Shell user aur kernel ke beech command interpreter/interface ka kaam karta hai.

Example: User command deta hai: Ls. Shell command ko samajhta hai aur OS se required kaam karwata hai.

Linux mein examples: Bash, Zsh, Fish.

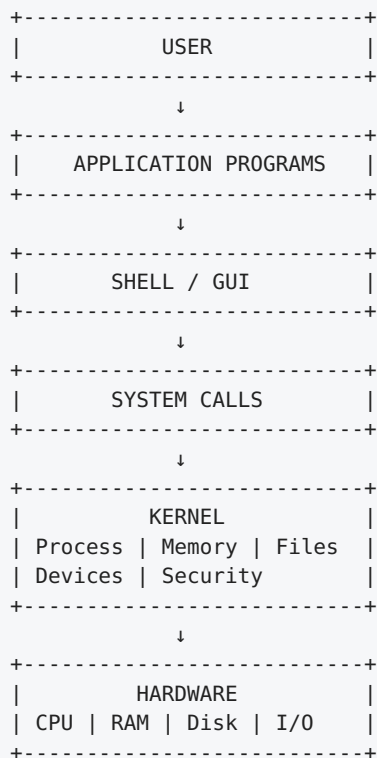
⑤ Application Programs

Ye wo programs hain jo user actual kaam ke liye use karta hai.

Example: Text editor, Browser, Media player, Compiler.

★ Linux Architecture Exam Diagram

Paper mein ye diagram bana dena:



6 System Call ★★★★★

Ye question pakka achhe se kar lena.

System Call kya hai?

System call ek mechanism/interface hai jiske through user program Operating System ke kernel se service request karta hai.

Simple language: Program ko OS se koi kaam karwana hai → System Call ke through request karega.

Example: Program ko file open karni hai.

System Calls ke Types

① Process Control

Processes ko create, execute aur terminate karne ke liye.

Examples: `fork()`, `exec()`, `exit()`, `wait()`

② File Management

Files ke operations ke liye.

Examples: `open()`, `read()`, `write()`, `close()`

③ Device Management

Devices ko access/control karne ke liye.

④ Information Maintenance

System/process ki information obtain ya modify karne ke liye.

Example: System time etc.

⑤ Communication

Processes ke beech communication ke liye.

Example: Pipes, Shared memory.

⑥ Protection

Resources ko protect karne aur permissions manage karne ke liye.

7 System Programs

Ab System Call aur System Program ko confuse mat karna.

System Program kya hai?

System programs aise programs hote hain jo user ko OS environment mein convenient way se kaam karne ki facility dete hain.

Simple:

System Call = OS se service maangne ka mechanism

System Program = user ke kaam ko easy banane wala program

Examples of System Programs

① File Management Programs

Files ke saath kaam karne ke liye.

Example: `ls`, `cp`, `mv`, `rm`

② Text Editors

Text/document edit karne ke liye.

Example: vi, nano

③ Programming Tools

Programming ke liye.

Example: Compiler, Assembler, Debugger

④ Communication Programs

Communication/networking ke liye.

Example: SSH, mail utilities

⑤ System Status Programs

System ki information dekhne ke liye.

Example: ps, top

🔥 System Call vs System Program

System Call	System Program
Kernel ko service request karta hai	User ko facility provide karta hai
OS aur program ke beech interface	User-level utility/program
Example: open(), read()	Example: ls, cp, vi
Usually programming interface	User directly use kar sakta hai

Ek Real-Life Example se sab connect kar

Maan le tu Linux mein ek file open karna chahta hai.



Yahan:

- **OS** → poora management system
- **Kernel** → OS ka core
- **System Call** → program ki kernel se request
- **System Program** → user ko useful tools dene wale programs
- **Shell** → commands ko interpret karta hai